

# Group Work Networking - Lab 1

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## Definitions of CIA

**Confidentiality:** Preserving authorized restrictions on information access and disclosure. How was hidden information violated?

**Integrity:** Guarding against improper information modification or destruction. How was sensitive information changed/modified?

**Availability:** Ensuring timely and reliable access to and use of information. How was access to sensitive information affected?

## Case Studies

### Case A — The 2017 Equifax Data Breach

What happened: A massive data breach at one of the largest credit bureaus in the United States. Attackers gained unauthorized access to the personal and financial data of nearly 150 million people.

**Confidentiality** - Through this attack, confidentiality was violated through around 147.9 million Americans, UK and Canadian citizens' personal information being exposed [1]. This information consisted of birth dates, addresses, social security numbers and driver's license numbers. Equifax reported that an additional 2.5 million American consumer records were accessed [1]. Sensitive personal information being publicly exposed showed a clear confidentiality breach.

**Integrity** - The violation of integrity was not directly evident due to the nature of the cybercrime, which was to steal sensitive data such as credit card numbers. For attackers to gain from this attack, they needed to ensure the legitimacy of their data so they could later sell it on the dark web [1]. There is no public information confirming the attackers tampered with or altered data.

**Availability** - The level at which data remained available was kept to a minimum, with the only instance being the Equifax online dispute portal being taken offline on July 30, 2017 [2], alongside other services taken offline as the company investigated and secured their systems.

#### Discussion Questions:

1. Primary Impact - The primary principle of violation would be confidentiality.
2. Justification - This cybercrime is known as identity theft, the practice of using an individual's personal information for financial gain. The information gathered had to be accurate to sell and profit on the dark web.
3. Secondary Impacts - Integrity was not violated; however availability was minorly affected as Equifax servers had to be down for around 1-2 days for investigation.

Sources: [https://en.wikipedia.org/wiki/2017\\_Equifax\\_data\\_breach](https://en.wikipedia.org/wiki/2017_Equifax_data_breach) [1] | <https://www.gao.gov/assets/gao-18-559.pdf> [2]

### Case B — The Stuxnet Worm

What happened: A highly sophisticated computer worm designed not just to steal information, but to cause physical damage. It specifically targeted the control systems of Iranian nuclear centrifuges, causing them to subtly malfunction and destroy themselves.

**Confidentiality** - With this case, confidentiality was violated as Stuxnet aimed to gather information on Iran's industrial systems and how they worked [3]. Stuxnet then used this information to cause

fast-spinning centrifuges to tear themselves apart [3].

**Integrity** - Stuxnet was primarily designed to destroy the centrifuges Iran used to enrich uranium as part of its nuclear program [4]. This affected integrity as Stuxnet altered the logic on PLCs responsible for controlling uranium enrichment, altering rotation speeds of centrifuges to cause self-damage and halt to the Natanz Nuclear Facility.

**Availability** - The availability of Iran's nuclear program significantly reduced following this attack, with one analyst estimating the program may be set back at least two years [4]. Around 2,000 centrifuges were rendered inoperable [4]. One official stated it was anticipated they would root out the virus within two months, however due to new unexpected versions appearing, this timeframe no longer seemed feasible [3].

#### **Discussion Questions:**

4. Primary Impact - The most significant impact would arguably be integrity.

5. Justification - The main goal was to destroy Iran's nuclear industrial facility. Through integrity being impacted, it stopped the entire operation of the facility, with confidentiality and availability both being secondary effects.

6. Secondary Impacts - Stuxnet also had minor confidentiality violations to execute the operation since the malware required insight into how the facility worked. Availability was also affected as it caused real-world physical damage.

Sources: [https://en.wikipedia.org/wiki/Stuxnet#Affected\\_countries](https://en.wikipedia.org/wiki/Stuxnet#Affected_countries) [3] |

<https://www.csoonline.com/article/562691/stuxnet-explained-the-first-known-cyberweapon.html> [4]

### **Case C — The 2016 Dyn DNS DDoS Attack**

What happened: A massive Denial of Service attack that targeted a major internet infrastructure company called Dyn. The attack flooded Dyn's servers with traffic, making many major websites such as Twitter, Spotify, and Reddit unavailable for hours across Europe and North America.

**Confidentiality** - In this case, confidentiality was not directly affected as users' data was not being compromised or accessed by unauthorized parties. Rather it was a case of reputational damage and outage of services relying on Dyn.

**Integrity** - Mirai malware causes devices to be remotely controlled and act like bot devices. This case concerns integrity as numerous IoT devices, including IP cameras and home routers, were being remotely controlled due to malware infection, resulting in botnets overwhelming the company's infrastructure with DNS lookup requests.

**Availability** - This attack caused major disruptions to popular websites and services such as Twitter, Netflix, Reddit, and Spotify. Users were unable to access them for extended periods. This resulted in one of the most significant DDoS incidents in history, illustrating how coordinated cyberattacks leveraging IoT vulnerabilities can cause widespread disruption.

#### **Discussion Questions:**

7. Primary Impact - The main impact was on availability.

8. Justification - Attackers flooded Dyn's servers with massive amounts of DNS lookup requests using botnet devices, overwhelming the company's infrastructure and causing massive disruption to popular services.

9. Secondary Impacts - This attack was not directly concerned about the integrity or confidentiality of user data. It was volumetric in nature with the main purpose of overwhelming servers with junk traffic.

Sources: [https://en.wikipedia.org/wiki/DDoS\\_attacks\\_on\\_Dyn](https://en.wikipedia.org/wiki/DDoS_attacks_on_Dyn)

### **Case D — The 2021 Colonial Pipeline Ransomware Attack**

What happened: A ransomware attack forced the shutdown of the largest fuel pipeline in the United States, which supplies nearly half of the East Coast's fuel. The shutdown lasted several days, leading to widespread fuel shortages and panic buying.

**Confidentiality** - Hackers gained unauthorized access to the pipeline system using a compromised password of an inactive VPN account without multi-factor authentication. Attackers stole around 100 gigabytes of data and threatened to make it public if ransom was not paid.

**Integrity** - The attack exposed serious integrity issues. A key failure was a vulnerable VPN account without multi-factor authentication. Investigation also revealed inadequate segmentation between the company's corporate IT network and its Operational Technology network responsible for the physical flow of fuel.

**Availability** - The six-day outage caused a severe fuel shortage and long lines at gas stations, resulting in a declaration of a state of emergency by the U.S. government. This disruption triggered a spike in gasoline prices and affected fuel-dependent industries. The company paid a ransom of \$4.4 million for a decryption tool, but it was reportedly too slow and they had to rely on their own backups.

#### **Discussion Questions:**

10. Primary Impact - The significant impact in this attack was on availability.

11. Justification - The company shut down its 5,500-mile pipeline to prevent malware from moving from the compromised IT network to the OT systems controlling the pipeline's physical flow. This closure lasted six days.

12. Secondary Impacts - Were on confidentiality and integrity. Hackers stole 100 gigabytes of data including information of nearly 6,000 employees, putting those individuals at risk of identity theft.

Sources: <https://www.bbc.co.uk/news/technology-57063636> |

[https://en.wikipedia.org/wiki/Colonial\\_Pipeline\\_ransomware\\_attack](https://en.wikipedia.org/wiki/Colonial_Pipeline_ransomware_attack)

### **Case E — The 2023 MOVEit Supply Chain Attack**

What happened: A Russian-affiliated cybercriminal group exploited a zero-day vulnerability in a popular file transfer software called MOVEit, stealing massive amounts of sensitive data from thousands of organizations worldwide including government agencies, financial institutions, and major corporations.

**Confidentiality** - Very much affected as hackers accessed and extracted financial and personal data including names, addresses, and national insurance or social security numbers, later threatening to release this data publicly if ransom was not fulfilled.

**Integrity** - There is no clear evidence of altering or modifying the compromised data. However, the integrity of the MOVEit software itself was compromised due to injection of malicious code, resulting in reliability concerns.

**Availability** - Affected moderately as MOVEit systems were temporarily taken offline by Progress Software and affected organizations to apply patches and investigate the breach, resulting in short-term operational disruption in data transfers.

#### **Discussion Questions:**

13. Primary Impact - Confidentiality.

14. Justification - Millions of records from both private and public organizations were compromised, highlighting the main goal of the attack was data theft and extortion.

15. Secondary Impacts - Integrity and availability were affected less than confidentiality due to service disruption and loss of trust.

Sources: <https://en.wikipedia.org/wiki/MOVEit> |

<https://www.csoonline.com/article/1248857/moveit-carnage-continues.html>

## Case F — The 2020 SolarWinds Supply Chain Attack

What happened: State-sponsored hackers compromised the software build process of IT management company SolarWinds, secretly inserting malicious code into a legitimate software update for the Orion platform, distributed to over 18,000 customers.

**Confidentiality** - Severely affected as attackers gained access to sensitive internal communications, network configurations, and classified government data. The attack allowed breaches to steal credentials and abuse confidential information over several months without detection.

**Integrity** - Even though there was no widespread evidence of the attackers altering customer data, malicious code was injected into the software updates causing serious concerns and trust issues within the software supply chain.

**Availability** - Once the breach was found, organisations had to take their SolarWinds systems offline, causing temporary disruptions to operations and IT management systems. The disruptions were not intended to be long-term as the attack was mainly to spy and not deny service.

### Discussion Questions:

16. Primary Impact - Confidentiality.

17. Justification - The SolarWinds attack primarily targeted confidentiality because the attackers' main goal was espionage, not destruction. Hackers gained unauthorized access to sensitive data including internal communications, government documents, and credentials over several months without detection.

18. Secondary Impacts - Integrity and availability were secondary impacts since SolarWinds attack is a classic example of a sophisticated supply chain compromise aimed at data theft and long-term surveillance.

Sources:

<https://www.techtarget.com/whatis/feature/SolarWinds-hack-explained-Everything-you-need-to-know>